Experiment 8

Report Scheduling, Automation, and Alerts using Microsoft Power BI

Aim

To design and implement report scheduling, automation, and alert mechanisms using Microsoft Power BI and develop strategies for automated report delivery, monitoring, and business performance tracking.

Tools / Software Required

1. Microsoft Power BI Desktop
2. Microsoft Power BI Service
3. Global Superstore Dataset.xlsx
4. Microsoft Excel

Theory

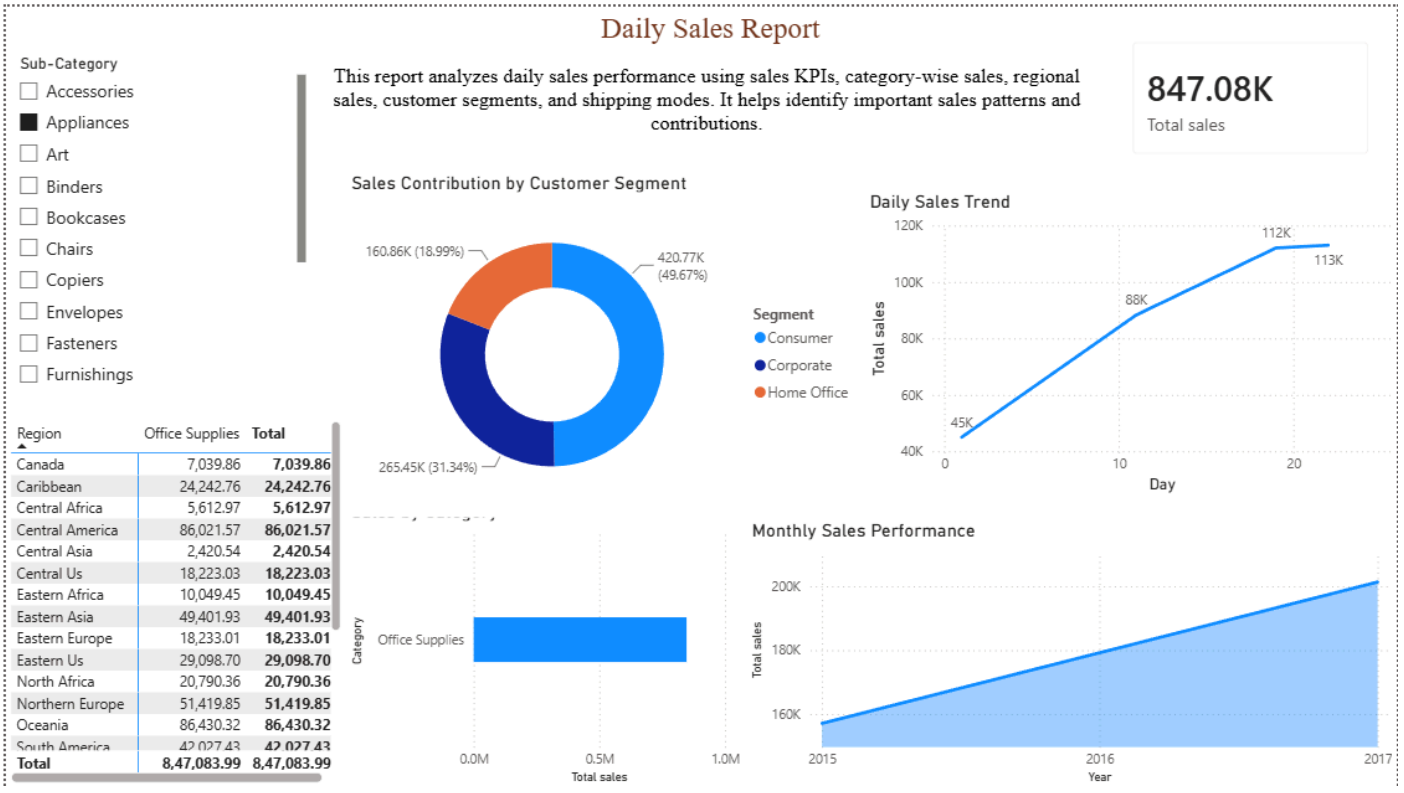
Business Intelligence systems must deliver information to stakeholders at the right time. Scheduled reporting, automated distribution, and KPI alerts help organizations monitor performance and take timely actions. Power BI Service supports subscriptions, scheduled refreshes, report sharing, and data-driven alerts.

Business Scenario

Global Superstore management requires automated delivery of reports and timely notifications whenever business performance changes significantly. Students will act as BI Operations Engineers and design a reporting automation strategy.

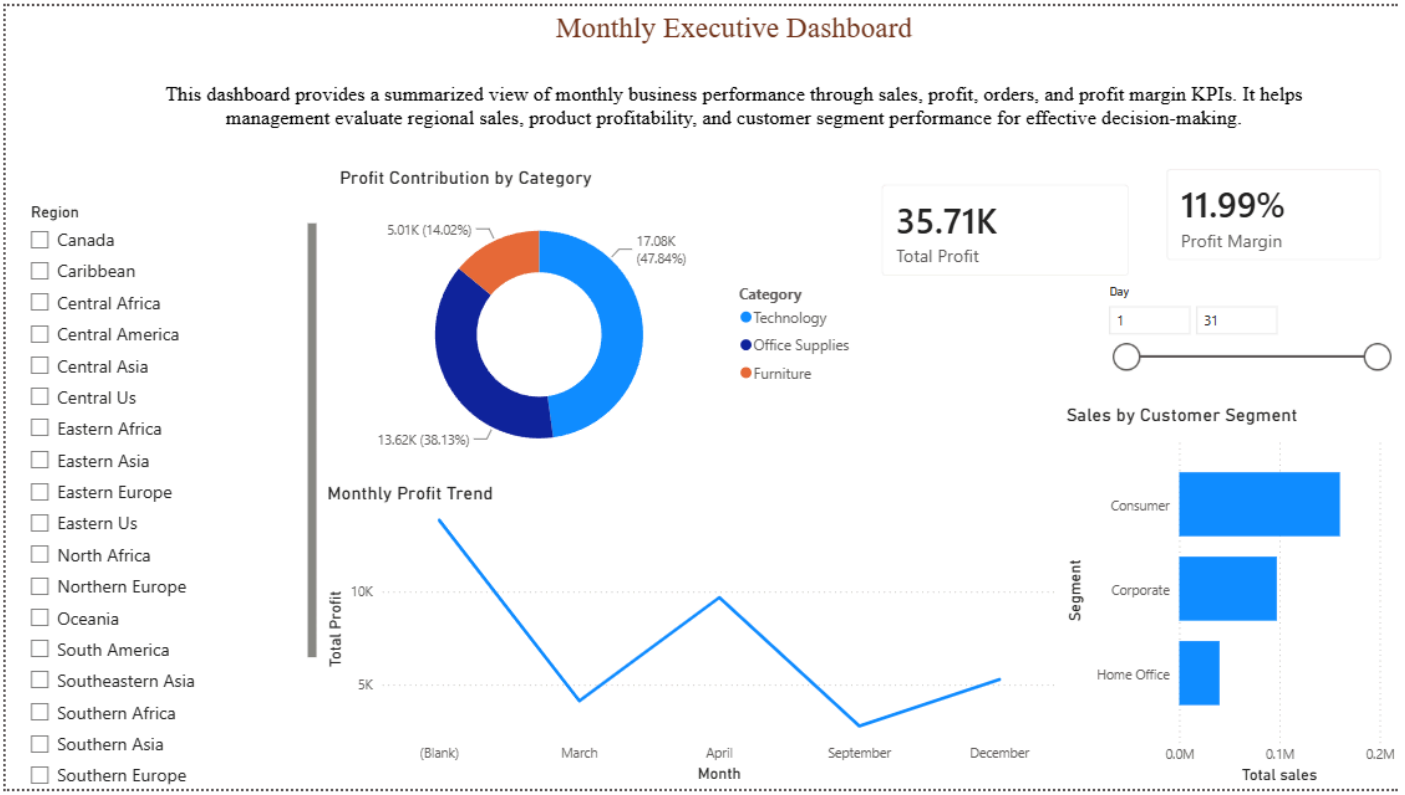
Conduct - Part A: Scheduled Reporting Plan

Create a scheduling plan for Daily Sales Reports, Weekly Profit Reports, and Monthly Executive Dashboards. Identify intended users and report frequencies.



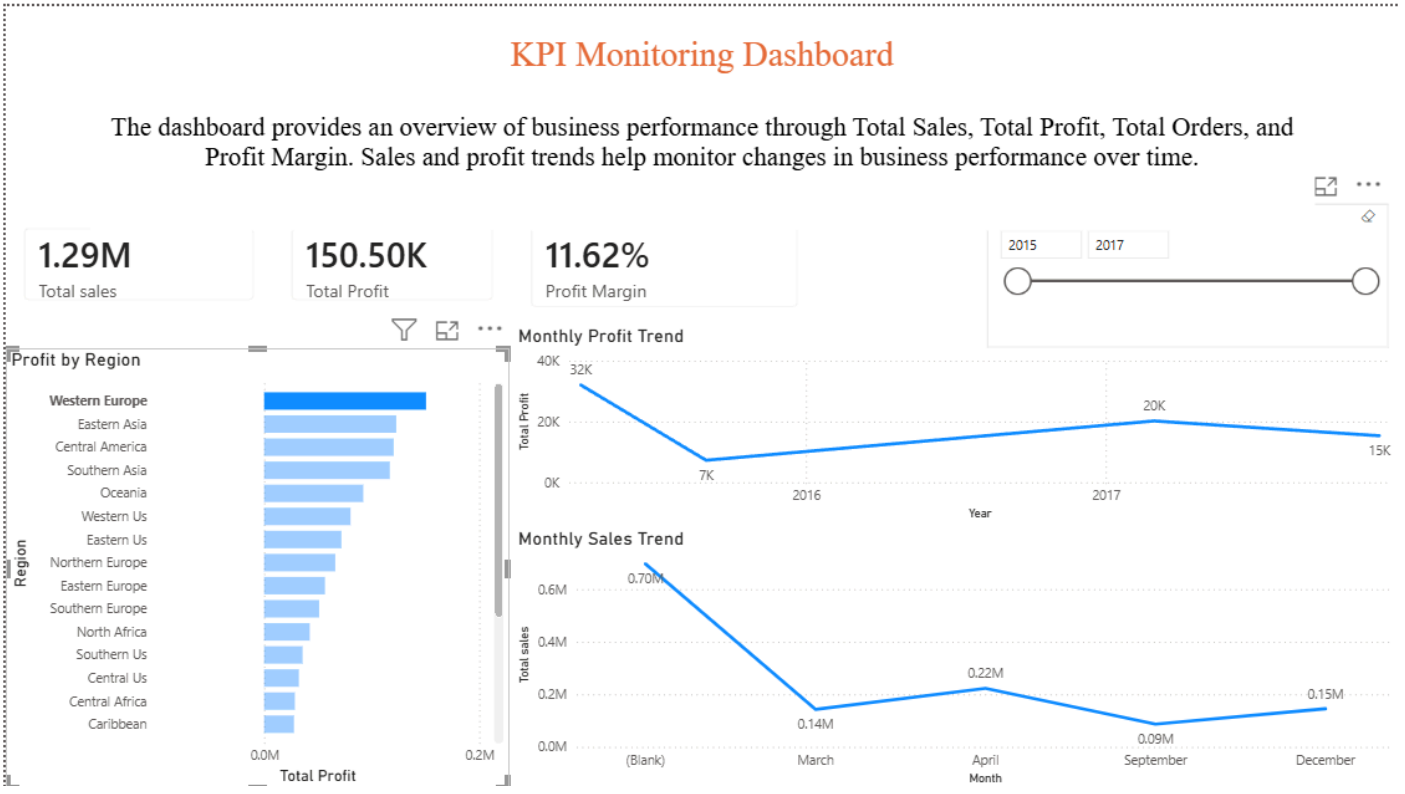
Conduct - Part B: Automated Distribution Design

Create a Report Distribution Matrix containing user groups, reports, frequency, and delivery methods such as Email, PDF, Shared Workspace, or Power BI Service.



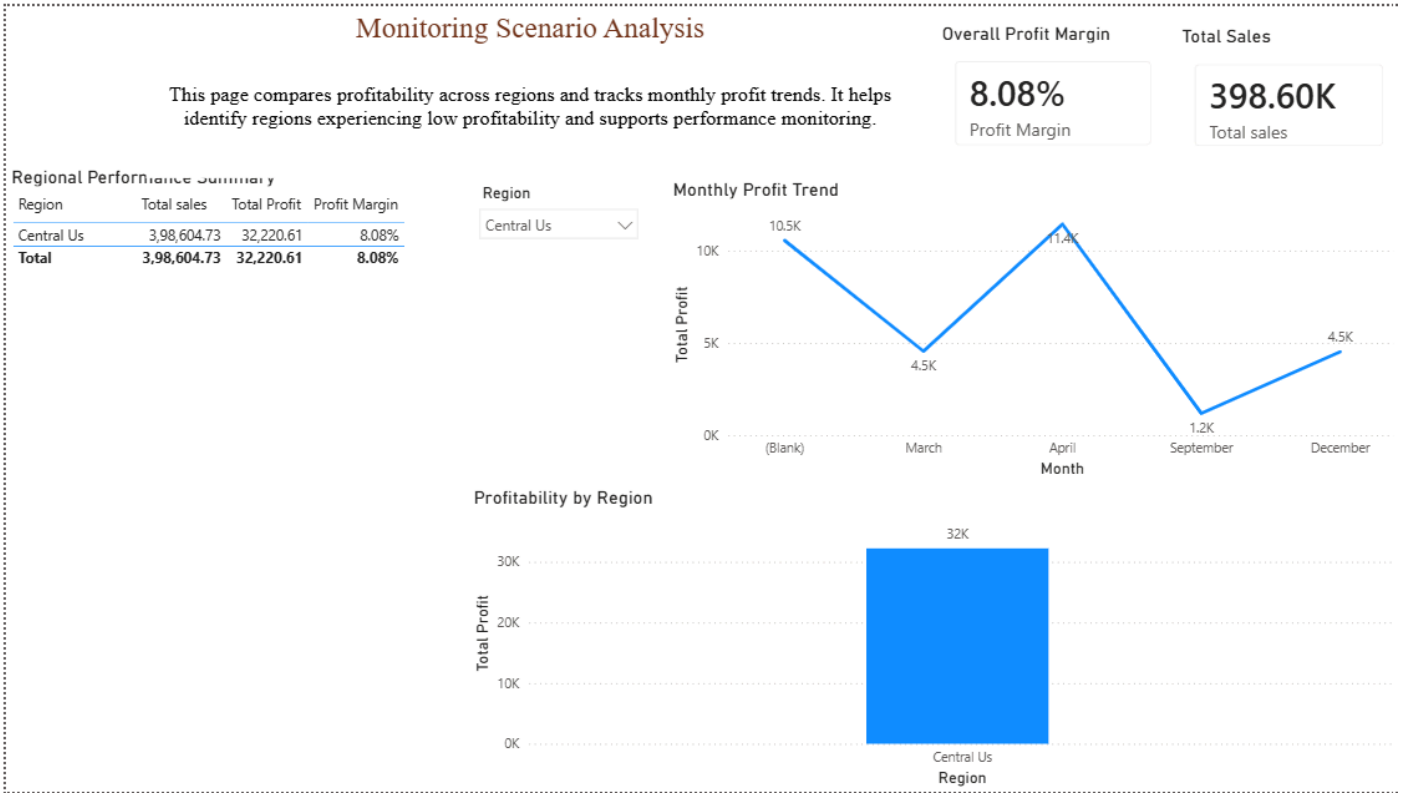
Conduct - Part C: KPI Alert Configuration

Define KPI alert conditions such as Profit Margin below 10%, Sales decline greater than 15%, and Returns increase greater than 20%. Specify recipients and actions.



Conduct - Part D: Monitoring Scenario Analysis

Analyze a scenario where a regional branch experiences declining profitability. Identify alert triggers, recipients, and corrective actions.



Deliverables

1. Scheduled Reporting Plan
2. Report Distribution Matrix
3. KPI Alert Configuration Sheet
4. Dashboard Screenshot
5. Monitoring and Recommendation Report

Result

Report scheduling, automation, and alert strategies were successfully designed. Automated reporting plans and monitoring mechanisms were developed to support business operations.

Learning Outcomes

- Apply report scheduling concepts in BI systems.
- Design automated report distribution strategies.
- Analyze KPI monitoring requirements.
- Evaluate alert mechanisms for business decision support.
- Develop recommendations for BI operations.

Text / Reference Books

1. Sharda, Delen & Turban, Business Intelligence and Analytics.
2. Cindi Howson, Successful Business Intelligence.
3. Rick Sherman, Business Intelligence Guidebook.
4. Microsoft Power BI Documentation.